

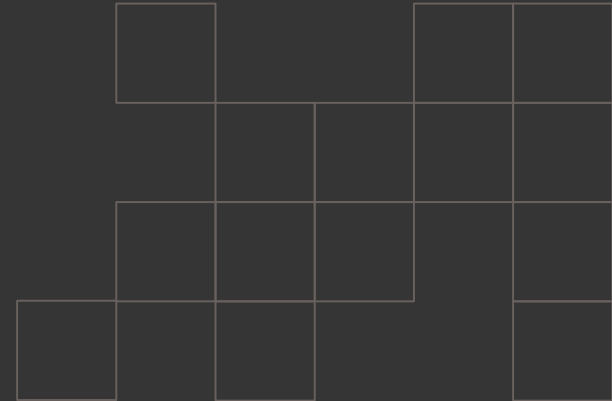
Scrum

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Credits to

Samin Sharaf Somik, Lecturer



Scrum



Source: Wikipedia

Scrum

- Scrum is a position in rugby in which players from two competing teams pack together closely to gain control of the rugby ball.
- In project management, however, scrum is a lightweight, iterative, and incremental agile process model.
- That means, scrum prioritizes simplicity in a project and develops product in a gradual manner with frequent delivery.
- Currently, scrum is one of the widely used process models in software industry.
- The word “**Scrum**” originally comes from **rugby**, where it refers to a **formation of players** who pack closely together with their heads down and attempt to gain possession of the ball.
- It symbolizes **teamwork, coordination, communication**, and **collective effort** — everyone pushes together toward a common goal.

Pillars of Scrum

Transparency

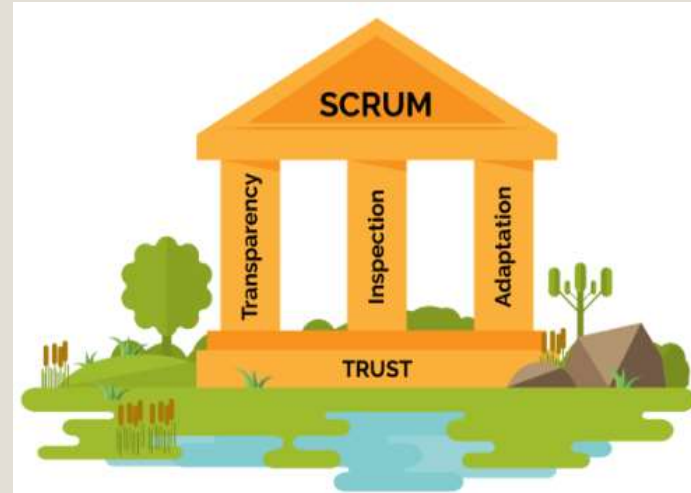
- All processes and progress must be visible to everyone involved.
- Artifacts like **backlogs** help maintain openness.

Inspection

- Regularly reviewing progress helps identify problems or deviations early.
- Inspections occur during meetings like **Daily Scrum** and **Sprint Review**.

Adaptation

- Based on inspection results, the team adjusts processes or goals to improve outcomes.
- Adaptation occurs in **Sprint Retrospectives** or when backlog priorities change.



Scrum Overview

Scrum Roles	Scrum Events	Scrum Artifacts
1. Scrum Master	1. Sprint	1. Product Backlog
2. Product Owner	2. Sprint Planning	2. Product Goal
3. Developer	3. Daily Scrum	3. Sprint Backlog
	4. Sprint Review	4. Sprint Goal
	5. Sprint Retrospective	5. Increment
		6. Definition of Done

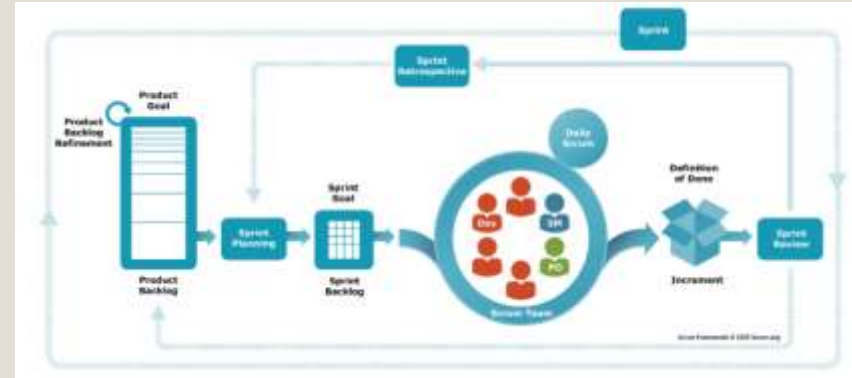
Sprint and Increment

Sprint

- The core timeboxed period (usually **1 month**) where a usable product increment is created.
- Has a fixed duration and consistent schedule.
- Each Sprint includes all other Scrum events.

Increment

- The **working product output** produced at the end of a Sprint.
- Must meet the **Definition of Done (DoD)** and be potentially shippable.

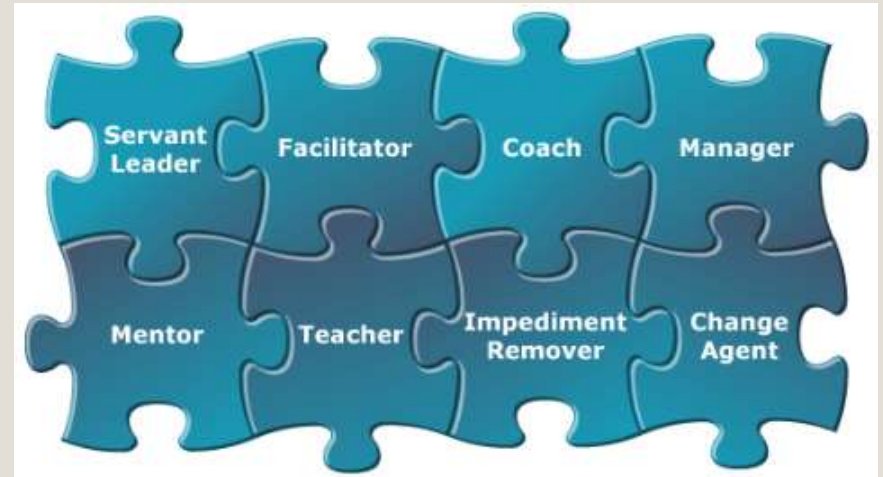


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Scrum Master

- Is responsible and accountable for the proper execution of **Scrum**
- Ensures the Scrum framework is properly **understood** and **followed**.
- Removes obstacles that hinder the team's progress.
- Acts as a facilitator, not a boss — supports collaboration and efficiency.
- Scrum master is usually the **project manager**.



Scrum Master

Helps the Scrum Team:

- By coaching the team members in self-management and cross-functionality
- Focus on creating high-value Increments that meet the Definition of Done
- Influence the removal of impediments to the Scrum Team's progress
- Ensure that all Scrum events take place and are positive, productive, and kept within the time-box.

Helps the Product Owner:

- Find techniques for effective Product Goal definition and Product Backlog management
- Provide ways for the Scrum Team to understand the need for clear and concise Product Backlog items
- Facilitate stakeholder collaboration as requested or needed

Supports the Organization:

- By Leading, training and coaching them in their Scrum adoption
- Remove barriers between stakeholders and Scrum Teams



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Product Owner

- Is responsible and accountable for the **quality of the product**.
- Represents the **customer or stakeholder** interests.
- Manages and prioritizes the **Product Backlog**.
- Defines clear goals and **ensures the team delivers maximum product value**.

Main Task:

- Developing and explicitly communicating the Product Goal
- Creating and clearly communicating Product Backlog Items
- Ordering Product Backlog Items
- Ensuring that the Product Backlog is transparent, visible and understood



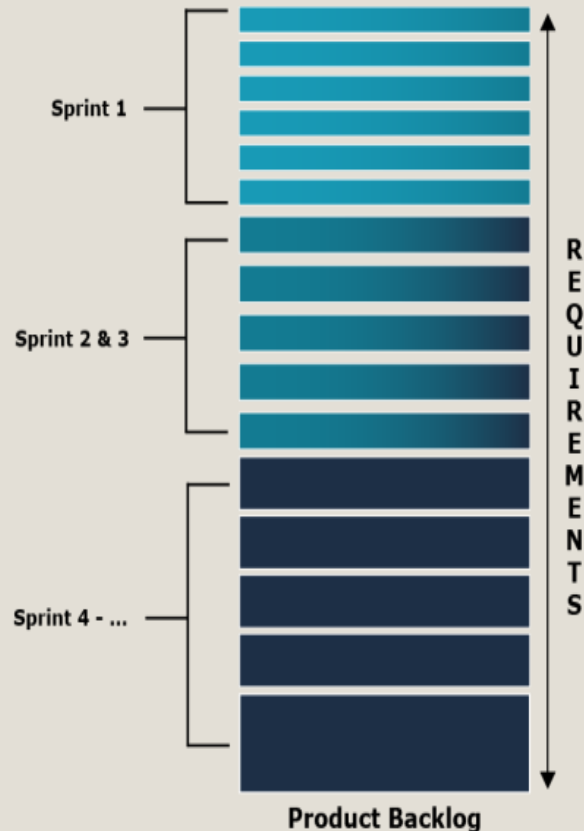
Product Backlog and Product Goal

Product Goal

- The **long-term objective** for the product.
- Provides direction for all Sprints and aligns the team's work.

Product Backlog

- An ordered list of **all desired features, fixes, and tasks** for the product.
- Continuously refined by the Product Owner.
- Commitment : **Product Goal**
- Product Backlog **refinement** is the act of breaking down and further defining Product Backlog items into smaller more precise items.

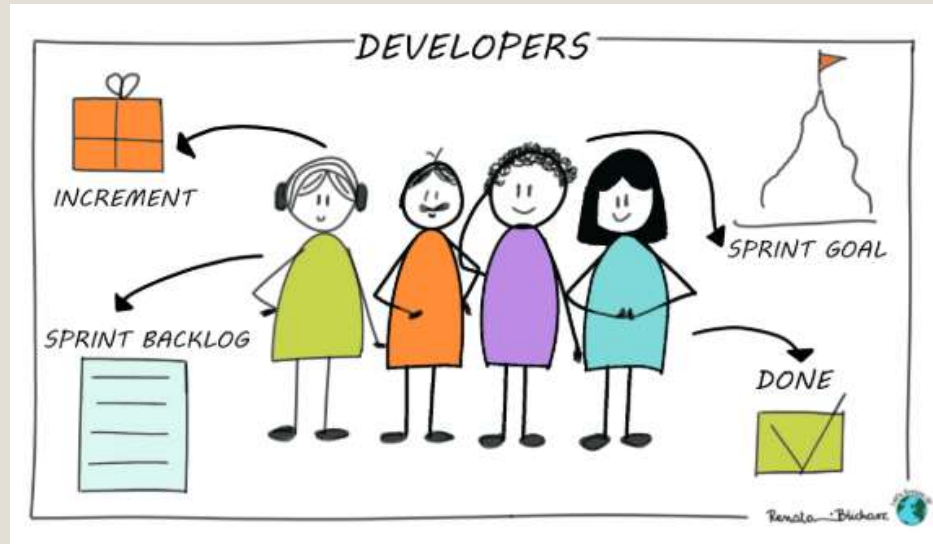


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Developers

- The professionals who **design, code, test, and deliver** the product increment.
- Responsible for achieving the **Sprint Goal**, the target of the Sprint.
- Create plan for the Sprint, the **Sprint Backlog**, and build an increment according to the plan
- Other roles: Facilitation, mentoring, teaching, coaching



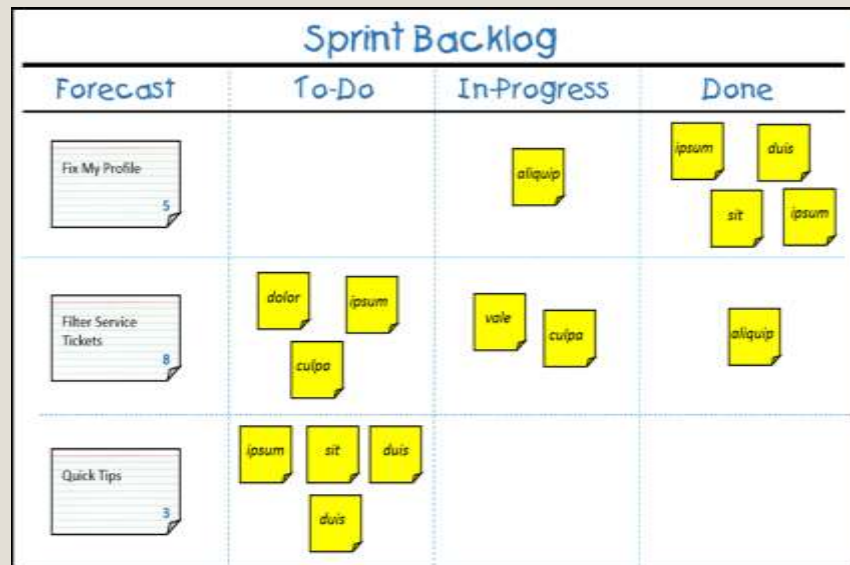
Sprint Backlog and Sprint Goal

Sprint Goal

- A **short-term objective or target** for the Sprint that guides the development team's focus.
- Set by the developers during **Sprint Planning**

Sprint Backlog

- A **subset of the Product Backlog** chosen for the current Sprint.
- Commitment : **Sprint Goal**
- Includes all tasks needed to achieve the Sprint Goal.



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Sprint Planning

- Held at the beginning of a Sprint.
- Defines **what** will be done (**Sprint Goal**) and **how** it will be achieved.
- The **Product Owner** presents the top-priority items, and the **Developers** select tasks.
- The **Sprint Backlog** is developed from the selected **Product Backlog** items.
- Timeboxed to a maximum of **8 hours**.



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Daily Scrum

- A short (**15-minute**) daily meeting for Developers.
- Team members discuss progress, upcoming work, and blockers.
- Sprint backlog items are discussed, and refined if necessary.
- Promotes coordination and keeps everyone aligned.



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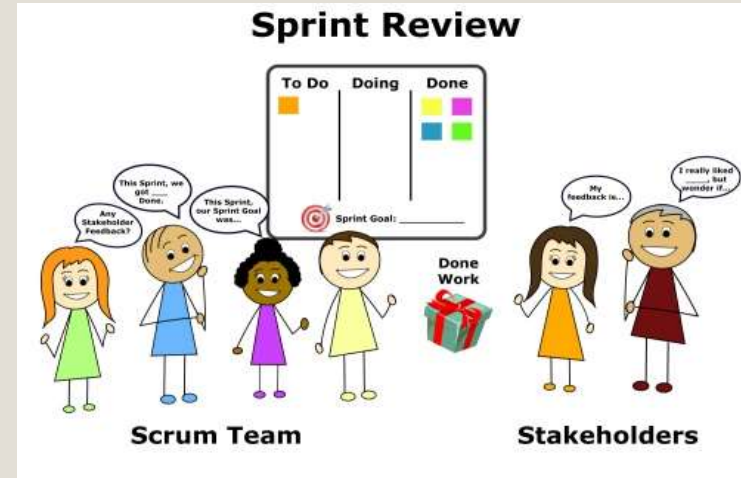
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Sprint Review



Sprint Review

- Held at the end of the Sprint to **inspect the increment**.
- The team demonstrates the new features to stakeholders.
- Stakeholders and Scrum team members are invited by the Product Owner
- Feedback is collected to improve the product backlog.
- Discusses which **Product Backlog** items are **Done** and which are **Not Done**.
- Discusses what problems the team faced, how they were resolved or not
- Review of product budget, timeline, marketplace demand, potential capabilities
- Timeboxed to a maximum of **4 hours**.

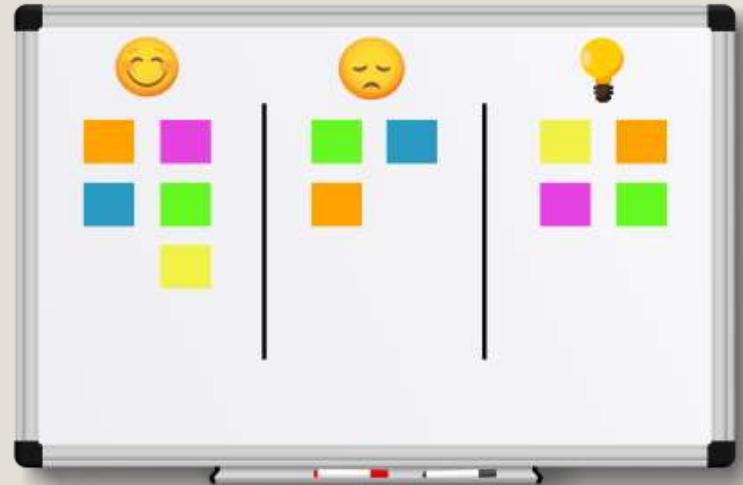


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Sprint Retrospective

- Final event of a Sprint.
- A short meeting of the Scrum Team after demonstration to the stakeholders.
- The team reflects on what went well, what didn't, and how to improve next Sprint.
- Timeboxed to a maximum of **3 hours**.
- Focuses on continuous improvement in teamwork and processes.
- Gives input for the next Sprint Planning.
- **Sprint Review** – Improves Product
Sprint Retrospective – Improves Process, Teamwork



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Definition of Done

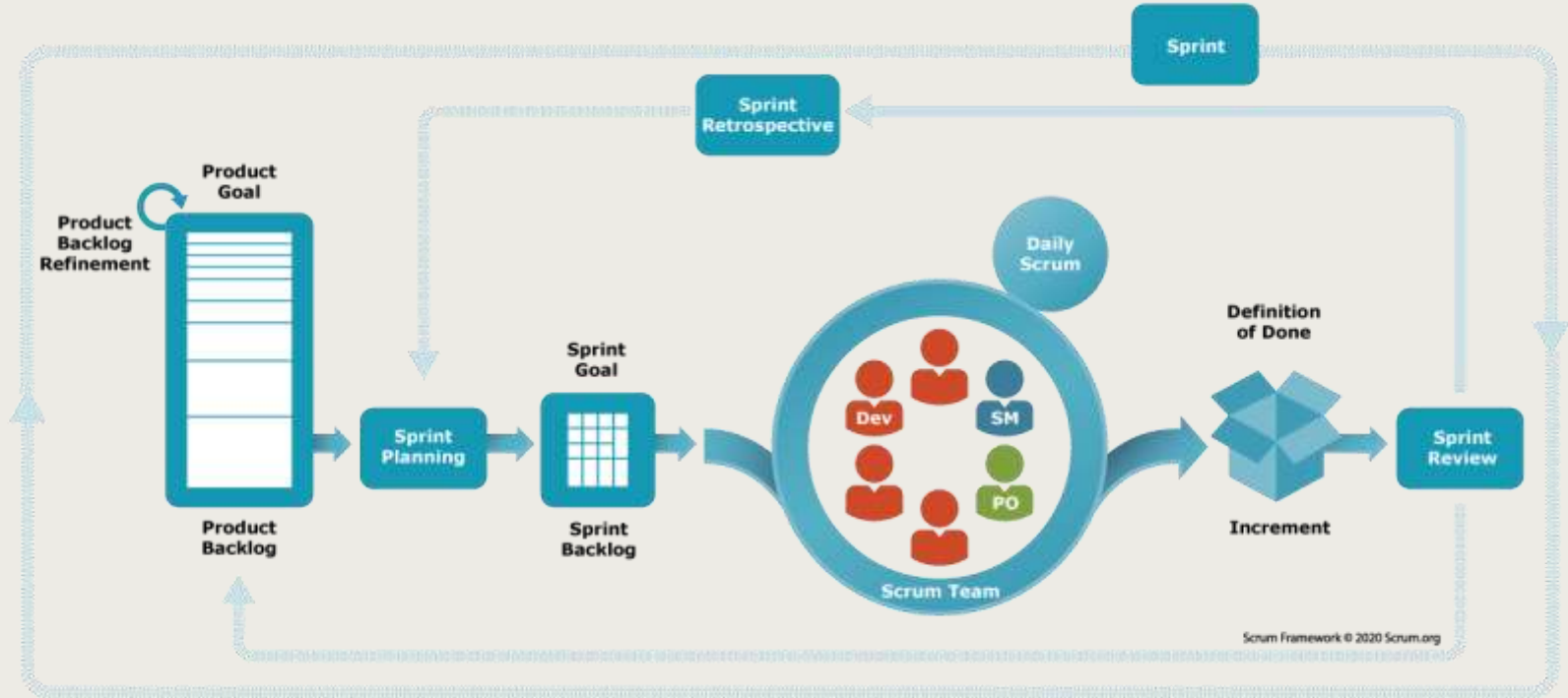
- A clear **checklist or standard** that defines when a product increment is considered complete.
- Ensures **quality** and **consistency** across Sprints.
- Commitment for the **Increment**.
- **Example:** DoD for a health-focused software application
 1. All testing completed
 2. No known defects
 3. Code review completed and passed
 4. Meets HIPAA Compliance standards
 5. Meets general security requirements



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Scrum



References

- Slides adapted from Md. Muhaiminul Islam Nafi, Lecturer, Dept. of CSE, UIU
- [What is Scrum? | Scrum.org](#)

Thank
You